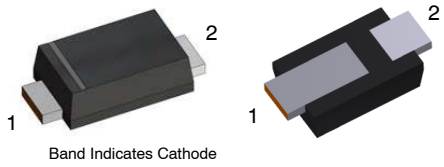
Surface Mount Schottky Barrier Rectifiers
1 A, 20 V - 150 V

SS12FP - S115FP

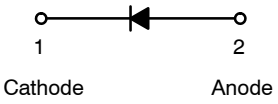


Band Indicates Cathode

SOD−123EP
CASE 425AC

Features

- Larger Cathode Pad for Improved Power Dissipation
- Ultra Thin Profile − Package Height < 1.0 mm
- High Surge Current Capability
- Low Power Loss, High Efficiency
- UL Flammability 94V−0 Classification
- MSL 1 per J−STD−020
- These Devices are Pb−Free and are RoHS Compliant

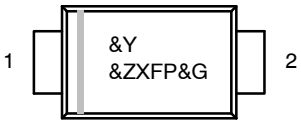


ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Value						Unit
		SS12 FP	SS13 FP	SS14 FP	SS16 FP	S110 FP	S115 FP	
V_{RRM}	Repetitive Peak Reverse Voltage	20	30	40	60	100	150	V
V_{RMS}	RMS Reverse Voltage	14	21	28	42	70	105	V
V_R	DC Blocking Voltage	20	30	40	60	100	150	V
$I_{F(AV)}$	Average Forward Rectified Current	1						A
I_{FSM}	Peak Forward Surge Current: 8.3 ms Single Half Sine−Wave Superimposed on Rated Load	30						A
T_J	Operating Junction Temperature Range	−55 to +125		−55 to +150				°C
T_{STG}	Storage Temperature Range	−55 to +150						°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

MARKING DIAGRAM



&Y = Binary Calendar Year Coding
&Z = Assembly Plant Code
XFP = Specific Device Code
X = 0, 2, 3, 4, 6, A
&G = Single Digit Week Code

ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

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THERMAL CHARACTERISTICS (T_A = 25°C unless otherwise noted) (Note 1)

Symbol	Parameter	Value	Unit
Ψ_{JL}	Thermal Characteristics, Junction−to−Lead (Note 2)	10	°C/W
$R_{\theta JA}$	Thermal Resistance, Junction−to−Ambient	140	°C/W

1. Per JESD51−3 recommended thermal test board. Device mounted on FR−4 PCB, board size = 76.2 mm x 114.3 mm.
2. Thermocouple soldered at cathode lead.

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Conditions	Value						Unit
			SS12 FP	SS13 FP	SS14 FP	SS16 FP	S110 FP	S115 FP	
V_F	Maximum Instantaneous Forward Voltage (Note 3)	I_F = 0.5 A			0.51	0.58	0.70	0.75	V
		I_F = 1.0 A	0.45	0.50	0.55	0.70	0.80	0.90	
I_R	Maximum Reverse Current at Rated V_R	T_J = 25°C	0.40				0.05		mA
		T_J = 125°C					0.50		
C_J	Typical Junction Capacitance	V_R = 4 V, f = 1 MHz	54				28		pF
Trr	Typical Reverse Recovery Time	I_F = 0.5 A, I_R = 1 A, I_{RR} = 0.25 A	6				14		ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

3. Pulse test with PW = 300 μs, 1% duty cycle.

ORDERING INFORMATION

Part Number	Device Code Marking	Package	Packing Method†
SS12FP	2FP	SOD−123EP	Tape and Reel
SS13FP	3FP	SOD−123EP	Tape and Reel
SS14FP	4FP	SOD−123EP	Tape and Reel
SS16FP	6FP	SOD−123EP	Tape and Reel
S110FP	0FP	SOD−123EP	Tape and Reel
S115FP	AFP	SOD−123EP	Tape and Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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TYPICAL PERFORMANCE CHARACTERISTICS

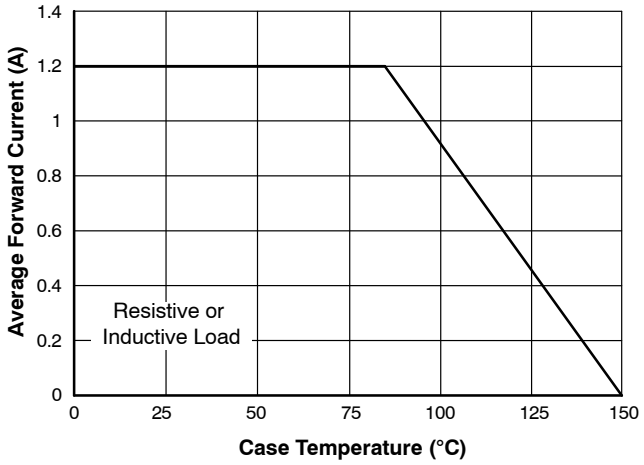


Figure 1. Forward Current Derating Curve

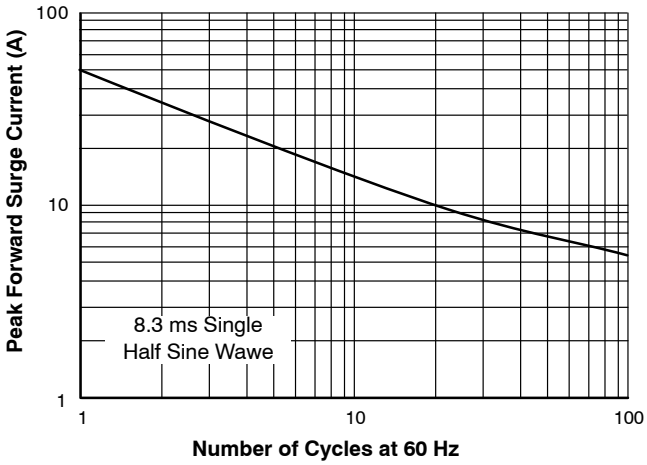


Figure 2. Maximum Non−Repetitive Forward Surge Current

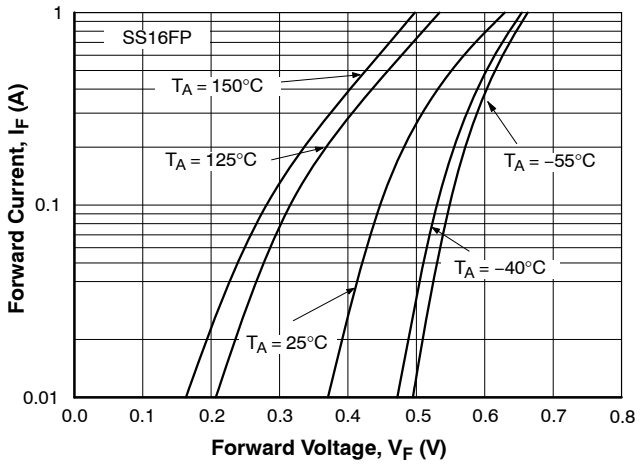


Figure 3. Typical Forward Characteristics

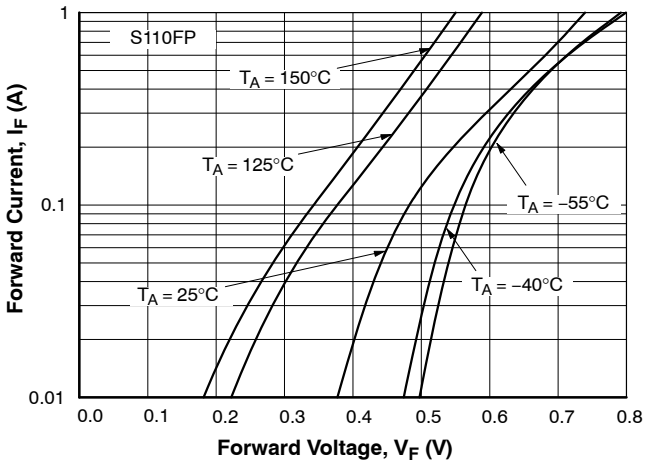


Figure 4. Typical Forward Characteristics

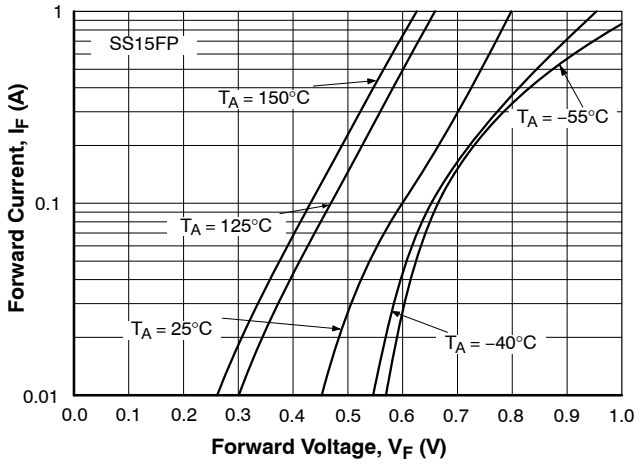


Figure 5. Typical Forward Characteristic

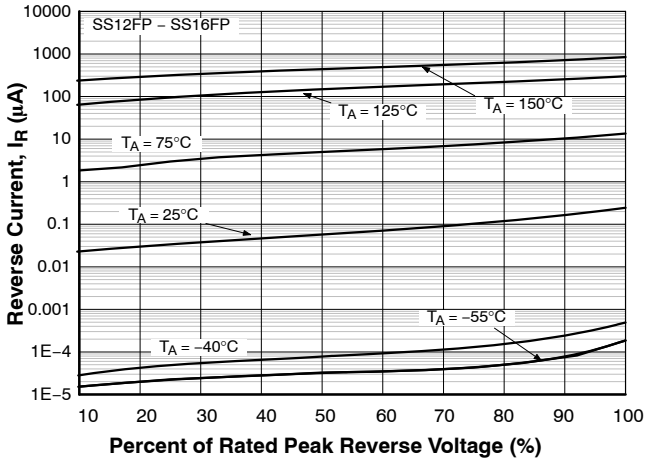


Figure 6. Typical Reverse Characteristics

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TYPICAL PERFORMANCE CHARACTERISTICS

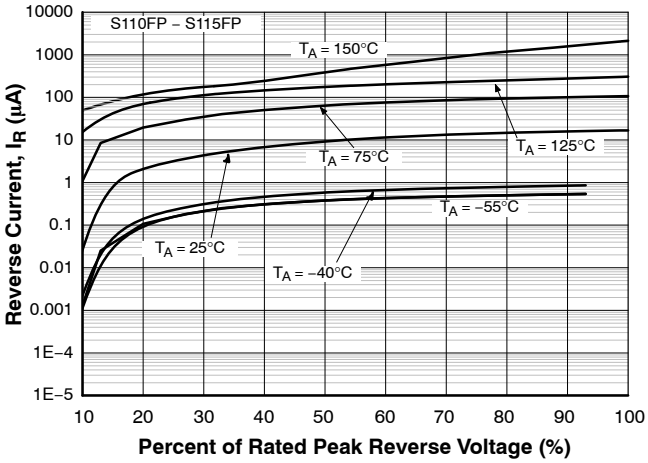


Figure 7. Typical Reverse Characteristic

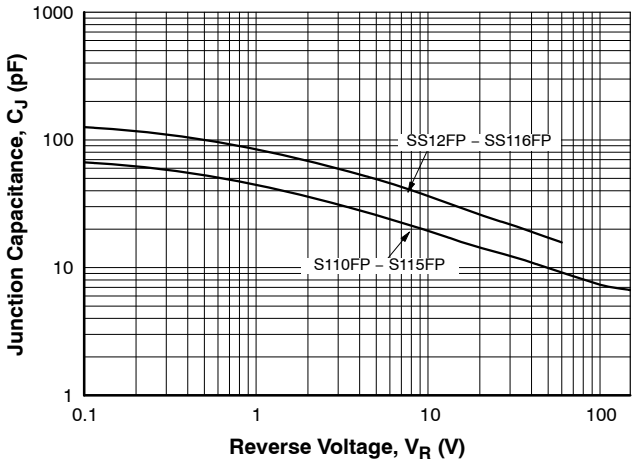
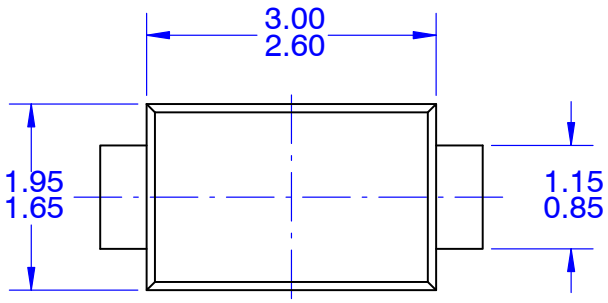


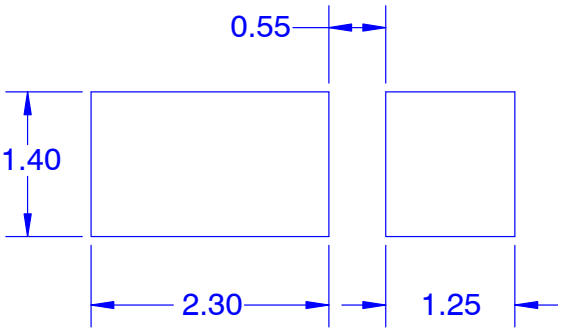
Figure 8. Typical Junction Capacitance

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SOD−123EP
CASE 425AC
ISSUE O

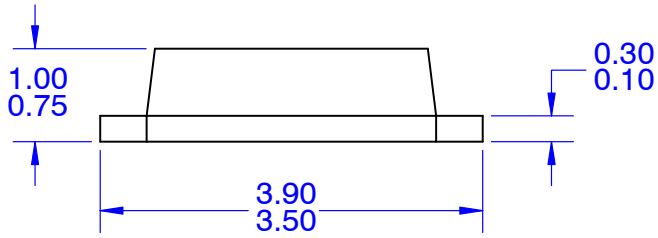
DATE 31 AUG 2016

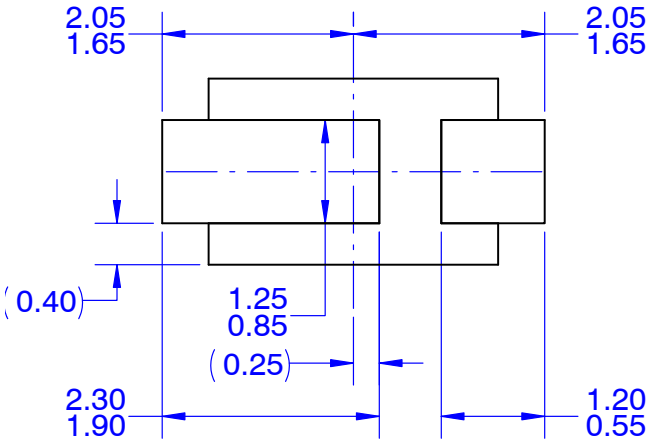




LAND PATTERN RECOMMENDATION

LONG PAD IS CATHODE





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